

Responsible AI usage for academic integrity in China

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Abstract

Purpose – This paper aims to explore how responsible artificial intelligence (AI) technology can enhance various aspects of academic achievement, benefiting institutions, teachers, students and society in creating productive and better educational systems.

Design/methodology/approach – Data were collected through interviews using convenience sampling involving students, university managers and lecturers. An inductive approach was applied among academic stakeholders in China.

Findings – The results show that students and lecturers use and emphasize AI technology in academics. Considering the responses, this study found that human-machine integration significantly improves the learning experience, including independent learning, better suggestions, ideas for solutions, module checking, curriculum setup and others. This study identifies the challenges associated with academic integrity and the unethical use of AI to enhance the learning experience and achieve better academic results.

Practical implications – The findings of this study help better understand the parameters that affect college students' and educators' confidence in and acceptance of AI systems. This paper has provided a theoretical foundation for developing AI systems that help students succeed academically and in various other learning contexts.

Originality/value – This paper has proposed a framework of responsible AI usage for academic integrity that can be used as the basis for understanding best practices. The study has identified the challenges and proposed a solution for the benefit of AI utilization in academia. AI technology is continuously developing, and its utilization will evolve in the future compared to how people are currently using it. In this study, the authors conclude that AI will certainly be widely used and could not be avoided. Educational stakeholders must adhere to clear standard guidelines and fairly assess AI and plagiarism-related work using expert human judgment, free from personal bias. As AI continues to advance, instructors and students will need to develop skills in identifying, evaluating, and investigating various aspects of AI in academic contexts.

Keywords Artificial intelligence, Learning experience, Academic integrity, Academic dishonesty, Moral awareness

Paper type Research paper

Introduction

Artificial intelligence (AI) is one of the most compelling aspects of how human existence has changed, even in education. AI, specifically computer systems, replicates human behavior and intelligence by machines (Telceken *et al.*, 2025; Leal Filho *et al.*, 2023). The current value of the worldwide AI market is \$142.3bn, and it is expected to continue to expand due to the influx of investments (Thormundsson, 2023). AI technologies have become increasingly prevalent in recent years, impacting academic performance and integrity. Incorporating AI and other technological innovations in educational settings has created several chances to enhance the learning experience, simplify administrative work and improve overall academic performance. According to Foroughi *et al.* (2023), AI will revolutionize learning and teaching in the future and will change the landscape of the

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educational system. However, other scholars have described the challenges of incorporating responsible AI into the educational system. The standardization of AI usage among academicians continues the debates (Akbar *et al.*, 2023; Ng *et al.*, 2023).

Along with the educational opportunities AI has brought forth, it also raises questions about students' adherence to academic honesty. The term academic integrity refers to the standards of conduct that should be upheld by all academic community members, including professors, researchers and students (Boehm *et al.*, 2009). Technology development and machine learning have completely changed educational environments and how students access information and acquire knowledge. Advanced information technology usage is supposed not to limit the exploration, innovation and integrity in teaching, learning and research, yet it will open new opportunities when it is used responsibly.

According to previous scholars, AI offers a significant opportunity to advance and revolutionize future teaching and research (Alhumaid *et al.*, 2023). Artificially intelligent software such as Bard AI and ChatGPT converse with and understand human language, and they also produce content based on scholarly and student's input (Mondal *et al.*, 2023). On the other hand, the adoption of AI has been viewed as a threat by certain people or scholars because it may produce content, which means that students could potentially abuse it (Limna *et al.*, 2022). Explaining how AI affects college students' ethical behavior and academic performance is vital as educational technology becomes more widely used. Understanding the complex relationship between generative AI technology adoption and academic integrity compliance is essential for educational administrators, faculties and policymakers. Furthermore, exploring the potential drivers of responsible AI usage will shed light on how adopting advanced technology tools influences college students' academic achievement and adherence to academic integrity.

There are several previous findings related to AI that reflect on its impact. However, it is difficult to determine to what extent AI blends and complies with academic integrity, especially among college students, which impacts their performance (Chan, 2023). It is also challenging to evaluate how AI affects academic users' behavior, attitude, motivation for using it, academic integrity, academic achievements and the specific role institutes play in ensuring ethical standards in AI and educational integration. This study has contributed to thoroughly identifying these aspects.

This study has provided insights and a better understanding of the detailed aspects linked with AI integration in academics, which will help users save time, focus more on research topics and manage complexity. It will also help global universities understand the concept of integration, ethical follow-ups, their responsibilities and the setting of standard practices. Finally, it will benefit users in understanding the associated misconduct issues and how they can use AI reasonably. This study addresses two research questions:

- RQ1. What specific factors drive the responsible conceptualization and practice of AI in the educational sector and how can these be standardized?
- RQ2. How can the insights from university stakeholders be used to develop a comprehensive AI usage framework that addresses both ethical and practical challenges?

This study has a unique proportion that offers value to the literature and addresses research gaps from the previous studies. The contribution of this study is as follows:

- Nowadays, with the advances in machine learning techniques, generative AI is becoming a part of human life, including in the academic field. Cao *et al.* (2024) argued that AI can avoid human error and suggest that skilled professionals need to benefit from the combination of human and AI capabilities. However, there is little evidence in

the literature about how AI is best used. UNESCO has provided recommendations as a basis for developing AI in the education policy framework. However, the UNESCO guidance was published before advanced AI tools such as GPT 3.5 and 4 were introduced in the market. UNESCO guidance does not fully address the current challenges and opportunities of advanced AI for education ([Chan, 2023](#)). The absence of clear guidance makes AI a black swan without recognizing what factors drive how responsible AI can be conceptualized and practiced in the educational sector. This study has offered motivational and beneficial insights from university stakeholders who comply with university and global standards.

- There is no comprehensive standard, and making AI is something that needs to be avoided in the academic world. Suppose there is a global agreement on how to use AI globally. In that case, it will help increase the decision-making quality, efficiency and productivity in academics, including the learning, teaching, and research processes, as well as consultancy and community services. Ethical issues are always a black menace for academics when using AI to carry out tasks and the learning process in academia. [Khalifa and Albadawy \(2024\)](#) suggested that the usage of AI in academics needs to emphasize ethical and transparent usage. Some academics consider using AI to be a mistake and violate ethics in the academic world without any apparent reference. The use of AI and ethical issues is a subjective matter, considering that it is important for lecturers to analyze which content has academic value and human thinking compared to works written using AI.
- Scholars need to have the proper training to integrate AI tools that balance human insight with machine learning. This paper has responded to the call from [Khalifa and Albadawy \(2024\)](#) that further research should address the ethical consideration and emerging challenge of AI usage in academics. This study has proposed the AI usage framework and insights that can be of guidance to university stakeholders. Our study has collected data from Chinese participants where AI is an essential element to study Science, Technology, Engineering and Mathematics in China since primary and secondary schools. The teachers use AI to review key points from the previous lectures and use robots to obtain real-time feedback and assistance to assess student performance and enrich and refine the courses ([Xinhua, 2024](#)). Our findings seek to find a solution to overcome challenges in using responsible AI in education that previous studies neglected.

The current study proposes enablers that motivate college students to use AI and explores to what extent universities are responsible for its ethical use. The organization of this paper is as follows. The next section is a literature review, which reviews past findings to understand the concept under investigation. The subsequent section has discussed methods for data collection and analysis. The results section is presented in the following section. Finally, at the end of this paper, the limitations of the research and future research opportunities are presented.

Literature review

The integration of Industry 4.0 into educational practices has sparked discussions about the impact of AI on both ethical behavior and academic achievement. As AI becomes more prevalent in academia, several ethical considerations must be addressed to ensure its responsible and fair use. These concerns primarily revolve around data privacy and security, transparency and accountability and informed consent. Data privacy and security are critical when implementing AI systems in academia ([Zeb et al., 2025](#)). These systems rely heavily on vast amounts of data to function optimally. Universities must safeguard the privacy and security of student and faculty data, protecting sensitive information from unauthorized access and misuse. Transparency and accountability are also essential when

using AI technologies in decision-making processes (Rezaei *et al.*, 2024). Academic institutions must be transparent about the incorporation of AI systems and establish clear guidelines to uphold accountability for the outcomes generated by these systems.

Informed consent is another crucial aspect. Data collected and analyzed using AI technologies from students or faculty members requires informed consent (Hossain *et al.*, 2025; Kalfeli and Angeli, 2025). It is recommended that individuals obtain informed consent, ensuring they are aware of how their data are used and have the choice to opt-out if desired. The ethical utilization of AI-generated content within universities can be delineated into multiple interconnected, overlapping and successive phases, including creation, dissemination, usage and evaluation. Each phase plays a crucial role in shaping the ethical landscape within academic institutions. AI tools, such as ChatGPT and Bard AI, offer significant advantages to university students in knowledge acquisition and information retrieval (Ahmed *et al.*, 2024). These tools leverage advanced natural language processing (NLP) and machine learning algorithms to provide instant access to a vast pool of information and resources. They contribute to the development of essential skills such as problem-solving, creativity and adaptability, preparing students for the modern workforce (Elbanna and Armstrong, 2024). The integration of AI tools in academic settings can revolutionize learning, collaboration and academic success. AI tools can analyze individual learning patterns and preferences to tailor educational materials and resources to suit each student's specific needs (Gligorea *et al.*, 2023). In addition, AI-powered tools can create interactive simulations and immersive learning environments that engage students in hands-on learning experiences (Rapaka *et al.*, 2025).

Al Ka'bi (2023) asserts that AI enhances student engagement by offering personalized instructional experiences and automating administrative processes. This active engagement enriches the learning process, facilitating the discovery of results and potential outcomes. AI algorithms streamline the assessment process by automatically grading assignments, providing detailed feedback and identifying areas where students may need additional support. Engaged students are more likely to participate in classroom activities, collaborate with peers, ask questions and seek clarification. These actions improve knowledge retention, sharpen critical thinking abilities and enhance overall academic performance. AI-generated content is not a novel concept in academia; 95% of students are familiar with tools like ChatGPT, and a portion of them use it regularly (Stöhr *et al.*, 2024). Despite this widespread use, many students lack awareness of the ethical rules and guidelines set forth by universities.

Lo (2024a) observes that users who subscribe to the premium version of AI-generated content exhibit higher levels of AI literacy compared to nonsubscribers. This suggests that access to advanced AI tools may contribute to a deeper understanding of AI technologies (Hornberger *et al.*, 2025). However, in the realm of AI ethics, users encounter various challenges (Lo, 2024b). These challenges include issues related to transparency, bias and fairness, particularly among specific user groups. Transparency concerns revolve around understanding how AI algorithms make decisions and operate (Kovari, 2024), whereas bias and fairness issues pertain to ensuring that AI systems do not perpetuate or amplify discriminatory practices and treat all individuals equitably. Addressing these ethical challenges requires a comprehensive approach that includes developing and implementing clear guidelines for the ethical use of AI technologies and promoting transparency in AI systems. In addition, it involves mitigating bias through careful design and data practices and ensuring fairness in algorithmic outcomes across diverse user populations. By actively engaging with these challenges and implementing ethical frameworks, users can navigate the complexities of AI technology responsibly and ethically.

Vigdor *et al.* (2014) argued that students face crucial decisions about their future before graduation. Some students may opt to further their education by pursuing an advanced degree or exploring a different field of study. AI systems have the potential to offer valuable

guidance to users based on their academic performance and interests in their intended field of study. The transition from academia to the professional world is a key aspiration. AI can play a significant role in assisting students during this critical juncture by providing career tools that align their aspirations with the pathways needed to achieve them. AI can provide students with real-time data and insights into job market trends, salary ranges, demand for specific skills and potential career growth opportunities. AI algorithms can analyze a student's skills, interests and academic background to suggest suitable career paths that align with their strengths and aspirations ([Bankins et al., 2024](#)).

In addition, AI-powered career advisors can offer personalized guidance based on a college student's background, experience, location preferences, abilities and professional aspirations. This tailored approach equips students with customized study tracks and potential opportunities for career advancement and growth. According to [Hunter and Bowen \(2023\)](#), IBM's AI-driven Blue Matching system represents a promising job-matching service for the future. This system leverages AI to connect individuals with suitable job opportunities based on their skills, qualifications and preferences. For example, Elon University uses AI technology to assist students in reviewing past courses, applying acquired knowledge to course planning, receiving personalized feedback and engaging in tailored learning experiences ([Hannan and Liu, 2023](#)).

The integration of AI technology into the educational sector has led to a significant transformation, automating various processes and enhancing teaching and learning practices. AI has become a critical platform in education, driving improvements in personalized learning experiences and educational outcomes. [Pataranutaporn et al. \(2021\)](#) suggested that the integration of AI in the education system empowers instructors to customize content and activities to align with each student's strengths, learning pace and areas of improvement. This tailored approach maximizes student understanding, engagement and performance in e-learning environments. By leveraging AI technologies, educational institutions can create adaptive learning environments that cater to the diverse needs of students, enhance the quality of education delivery and foster a more engaging and practical learning experience. AI's role in education continues to evolve, offering innovative solutions to optimize teaching methodologies, student support and overall educational outcomes ([Kovari, 2024](#)).

[Seo et al. \(2021\)](#) propose that AI has the potential to revolutionize education, enhancing its accessibility, success and efficiency for students worldwide. AI not only aids students in understanding and learning but also provides real-time assistance to higher institutional students. AI-powered tutoring systems can identify specific difficulties or challenges that students encounter, enabling educators to offer targeted explanations, feedback and additional resources to assist students in overcoming these obstacles ([Pataranutaporn et al., 2021](#)). Moreover, AI can simplify the grading process for assessments like multiple-choice questions. Automating grading tasks enables educators to save time and establish a more effective feedback loop ([Sallam et al., 2024](#)). This loop allows students to monitor their progress and empowers them to make essential improvements in their educational path. Recognizing that the effective incorporation of AI technology in education necessitates deliberate design, ethical considerations and continual assessment is crucial. While AI holds significant promise, it cannot replace the role of human teachers. Instead, technology should be viewed as a supplementary tool to enhance and enrich the teaching and learning experience.

AI is not inherently frightening. When used with integrity and responsibility, it can enhance the effectiveness and performance of individuals and institutions. The relationship between academic achievement and AI in education is continuously evolving. It is set to undergo further transformation as AI technology progresses and becomes more widespread in the educational landscape. This paradigm shift presents educators and students with a spectrum of opportunities and challenges. Educators must adeptly adapt to leverage the

benefits of AI, navigating complexities while ensuring that the human touch in education remains paramount ([Abulibdeh et al., 2024](#)). The fusion of AI technology and education holds the potential to revolutionize conventional teaching methodologies, offering tailored learning experiences and elevating academic performance. By embracing AI as a supportive tool and recognizing its evolving role in education, educators can harness the transformative power of technology to cultivate a more effective and engaging learning environment for students in the years ahead.

Methods

This section discusses the procedure for collecting and analyzing data. To achieve the research objectives, our study used the inductive approach, which does not need strong support from preexisting concepts and theories ([Khan and Sarkar, 2024](#)). We have examined the role of responsible AI for academic integrity among Chinese participants using a qualitative approach to discover the in-depth results that have not been covered by quantitative research. According to [Mukumbang \(2023\)](#), the application of philosophies such as positivism, realism, interpretivism, and pragmatism is closely tied to the investigator's adoption of a specific research objective. In this study, an interpretivism approach has been embraced, placing emphasis on personal meanings and motives as the central focus of the research. By adopting an interpretivism approach, the study seeks to delve into the subjective interpretations and contextual understandings of individuals, acknowledging the significance of personal perspectives and motivations in shaping their experiences. This approach aligns with the idea that reality is socially constructed and that individuals actively contribute to the creation of meaning within their environments.

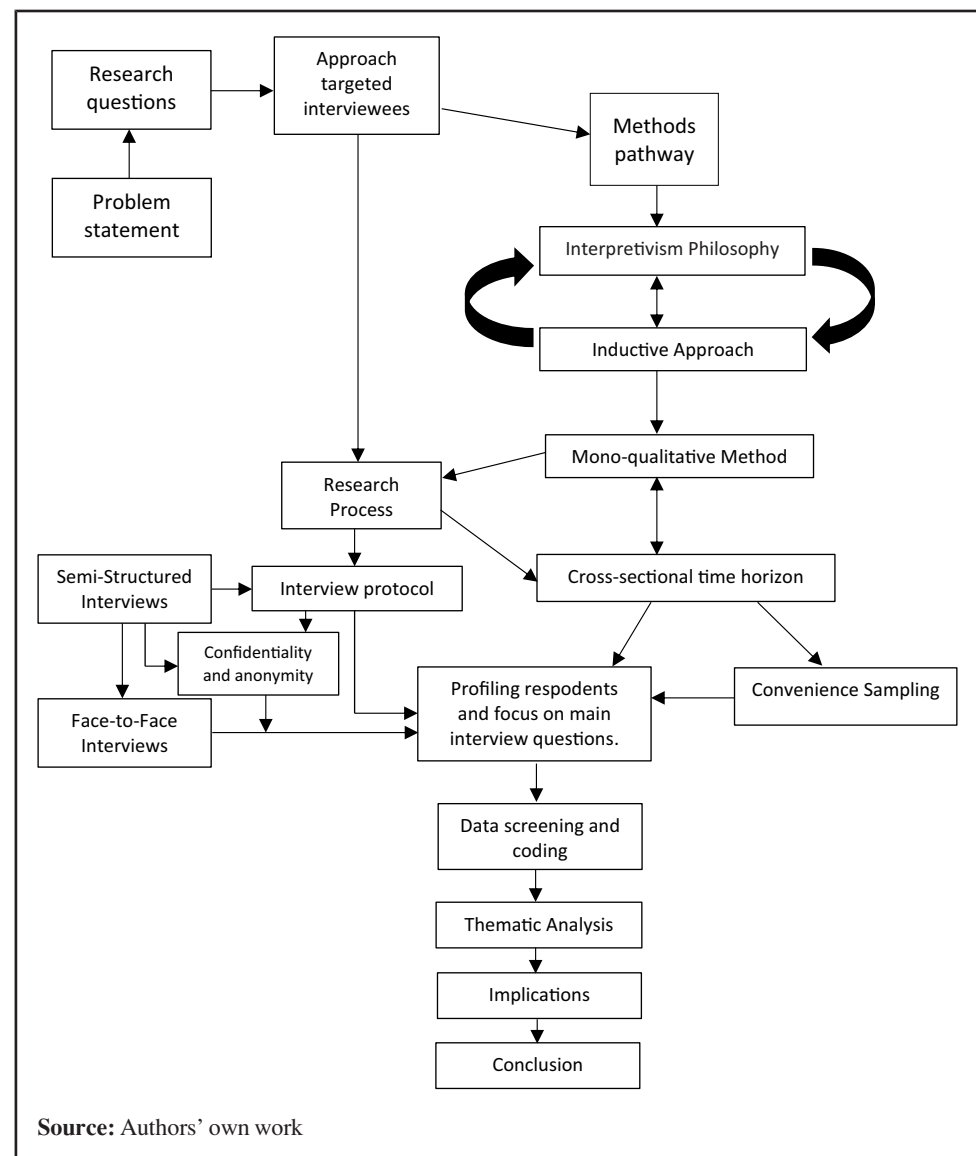
To gather rich and nuanced data that captures the intricacies of personal meanings and motives, potential respondents have been proactively invited to participate in the study. This proactive engagement strategy reflects a deliberate effort to involve individuals who can offer valuable insights and diverse perspectives, enhancing the depth and breadth of the research findings. Through this approach, the study seeks to not only understand the explicit experiences and viewpoints of participants but also to uncover underlying meanings and motivations that drive their actions and perceptions. By embracing interpretivism and actively engaging with respondents, the research aims to generate comprehensive insights that illuminate the complexities of personal interpretations and motivations within the context of the study.

Using a convenience sampling technique, this study aims to collect data from a readily available pool of participants efficiently. By using this method, authors can access a diverse range of respondents with relative ease. As part of the data collection process, respondents were queried about their previous encounters with AI within academic contexts. This initial filter question serves to gauge respondents' familiarity and experiences with AI, providing valuable insights into their perspectives and interactions with the technology in an educational setting. Verifying the information provided and ensuring consensus through cross-checking are crucial aspects of conducting a qualitative study ([Fernando et al., 2012](#)). To achieve the research objectives, a well-defined plan and process are essential, which the research technique offers. The inductive approach is used in this study, as no prior assumptions are made regarding the frequency of AI usage by respondents. The investigation analyzes information from respondents through thematic analysis ([Fernando et al., 2022](#)). In this study, the authors have utilized a mono-qualitative approach, aiming to capture descriptive responses that provide in-depth comprehension and minimize the potential for incomplete information often present in quantitative data. The type of data being collected will dictate the research strategy, providing a framework for using various data collection and analysis techniques ([Carter et al., 2023](#)). The authors have opted for a cross-sectional time horizon, as responses will be gathered within a limited time frame, allowing for a snapshot of perspectives at a specific point in time.

Figure 1 illustrates the research process. The research commences by identifying a clear problem statement concerning the ethical considerations of using responsible AI within academia. This study has focused on this issue and research gap as the cornerstone of the research endeavor. We have followed the interview procedures outlined in previous studies (Maher and Bedwei-Majdou, 2025; Sánchez-Guardiola Paredes *et al.*, 2021; Kallio *et al.*, 2016). The research process steps are as follows:

Based on the problem statement, two research questions have been formulated to direct the investigation. We have crafted clear and concise research questions: What specific factors drive the responsible conceptualization and practice of AI in the educational sector, and how can these be standardized? How can the insights from university stakeholders be used to develop a comprehensive AI usage framework that addresses both ethical and practical challenges? Following this, we identify the target interviewees who can offer valuable insights related to the research questions. We have identified profiles of the selected respondents, outlining their background, expertise and relevance to the research.

Figure 1 Research process



Factors such as expertise, experience and perspective are considered when selecting individuals to interview.

We provide a brief overview of the method used in the study. This includes outlining the overall research design, data collection methods and analysis techniques that will be used. Our study has developed a semi-structured interview protocol containing specific questions on the topic under investigation. This protocol ensures consistency across interviews and aids in gathering relevant data. We found that semi-structured interviews are suitable for collecting data due to their balance of structure and flexibility. This format allows interviewers to probe deeper based on participants' responses.

One profile question and ten interview questions were formulated, and their relevance was to answer the two research questions from the perspectives of students, lecturers and the manager. The interviews were conducted face-to-face and were convenient for the respondents and investigators. To ensure the transparency of the research, the respondents are requested to read the purpose of the study with a cover letter and given the interview questions three days in advance. We conducted the interviews in comfortable venues to ensure that interviewees felt at ease, which encouraged honest responses and reduced anxiety. Interviewers acted as active listeners, using probes to enable respondents to provide more detailed information. We informed participants of the ethical considerations, emphasizing that the information collected would be kept confidential and anonymous. The information gathered from the interviewees was recorded and supplemented with detailed note-taking to ensure accuracy and completeness.

To ensure the data validity, our study conducted data saturation and cross-checked the responses. After conducting interviews, the collected data needs to be screened for relevance and coded for analysis. We have compared responses across different participant groups or sources. The principle of saturation was to identify the potential mismatch between the interview questions designed for students and the participants who were university lecturers and a manager.

This process entails organizing and categorizing the data to pinpoint key themes and patterns. The authors have used thematic analysis to identify and interpret patterns within the data. This method involves systematically analyzing the coded data to reveal recurring themes, insights and connections that address the research questions. To ensure the validity and robustness of the results, we conducted a cross-case analysis based on the information provided by the respondents. Cross-case analysis aims to ensure the consistency of patterns and themes across different cases. The variation in answers stems from respondents belonging to other group categories. This variation enhances the robustness of the findings. After we have collected the information from interviewees on responsible AI and conducted the thematic analyses, we review and reflect on the key points and new insights.

We have discussed the implications of the research findings, highlighting how the insights gained from the study contribute to existing knowledge, practice or theory. The authors summarize the essential findings and suggest potential avenues for further research.

Results

Most of the respondents actively use AI to assist their work in universities (on average, two years of experience). The respondents agreed to participate and their identities remain anonymous. An online and recorded interview was conducted and took an average of 72min. There are a total of 10 participants, and the question size is 11, including profile questions. Ten participants were selected to find the consensus and validity of respondents. Additional respondents are required if the consensus and data cannot achieve saturation. All these can provide significantly vital information to meet the objectives of this study. The majority of respondents actively use AI to support their work within

universities, with an average of two years of experience. These respondents have voluntarily agreed to participate in the study, and their identities have been kept anonymous to ensure confidentiality and privacy.

The research process involved conducting online and recorded interviews with the participants, with each interview session lasting an average of 72 min. A total of 10 participants were selected for the study, and the interview questionnaire comprised 11 questions, including profile inquiries. By involving 10 participants, the goal was to seek consensus and validate the responses provided. Ten respondents were chosen as a sample size to ensure a diverse range of perspectives and experiences related to the use of AI in academic settings. Suppose saturation of data is not achieved, meaning that new insights and themes are no longer emerging from the interviews based on our data. In that case, additional respondents are not required as we have enough data and comprehensive coverage of the research topic to validate the findings. The information gathered from these participants, as well as their experiences, insights and perspectives, is crucial in answering the research questions. Our results play a significant role in providing valuable information and understanding the motivations, challenges and implications of using AI in academic environments, ultimately contributing to the overall success and impact of the research. The profiles of respondents and data linkage are presented in [Table 1](#). The ten interview questions listed in [Table 1](#) were designed from the perspective of students, university lecturers and managers as AI users. The interview questions are generic and can be applicable to three types of respondents.

By using appropriate data analysis techniques, the examination of the obtained data can yield the intended outcomes. Quantitative data can be analyzed using methods such as frequency analysis, regression and correlation, whereas qualitative information can be analyzed using narrative analysis, thematic analysis and content analysis ([Ott and Longnecker, 2015](#)). A thematic analysis was conducted in this study as the questions were descriptive, and creating themes would facilitate a better evaluation of the respondents' statements.

Transparency, privacy, confidentiality, secure data disposal and other ethical considerations are pertinent to primary research ([George, 2016](#)). Before collecting data, participants were given the opportunity to review the consent documents and participant information provided by the institution. To avoid potential bias during data collection, we have implemented several mechanisms. We follow the interview guidelines to ensure the same experience for all interviewees by defining the structure of the interview, including what questions to ask and in which order. We use the standard set of questions and semi-structured interview process similarly for each candidate. We ensure that our questions are clear, neutral and not leading. All data have been documented with secure password-protected, ensuring the confidentiality of the data throughout the study. A thematic analysis was conducted to categorize the results to meet the objectives effectively. When evaluating the results of the first question regarding demographic aspects, it was observed that the participants were not from a single region or work culture, and the majority were over the age of 40. In addition, the majority of respondents have postgraduate backgrounds. Some participants are lecturers, enhancing the research's validity from both student and lecturer perspectives. Two primary thematic analyses have been identified, and the details are as follows.

Theme 1: Motivation and benefits to use artificial intelligence in academics

The motivation and benefits of integrating AI in academic settings are multifaceted and offer a range of advantages to both educators and students. [Figure 2](#) shows the motivation and benefits of using AI. Evaluating the response for motivation to use AI, it is found that most of the respondents feel motivated to use it for optimizing personal learning evaluations, collecting and processing information, completing tasks on the teaching platform,

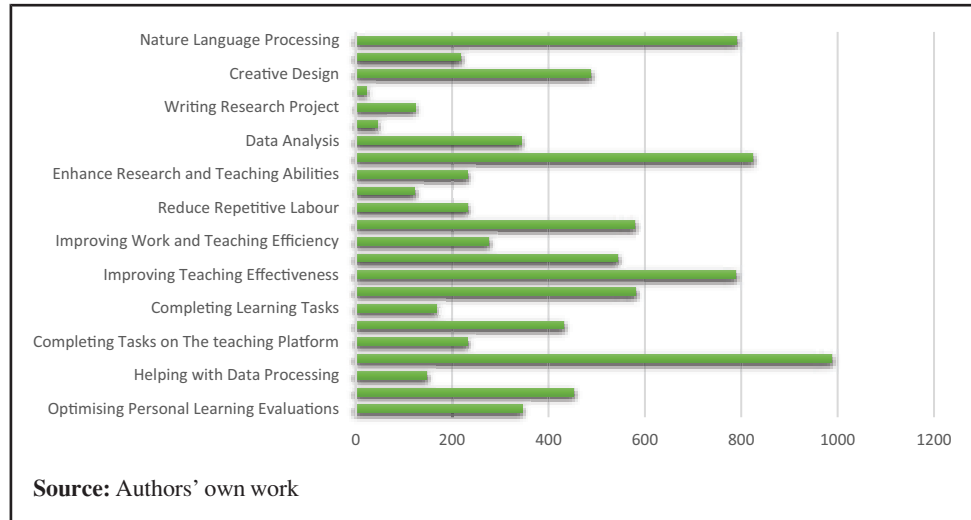
Table 1 Profile of respondents and data linkage

No.	Status	Profile Age	Education (location)	Research questions	Data linkage Interview questions
1.	Postgraduate student (the School of Engineering)	26	Bachelor (East China)	1.What are the users' motivations and benefits to use AI and comply with the academic standards set by the university?	1.Please provide your background information, including your job position, major and age
2.	Postgraduate student (the School of Engineering)	22	Bachelor (South China)	2.How to address the challenges associated with implementing AI policies in academics?	2.In which areas do you usually use artificial intelligence technology?
3.	Lecturer (Engineering and Technology)	36	Master (Western China)		3.What motivates you to use AI?
4.	Lecturer (Art and Design)	32	Master (East China)		4.What are the applications of AI technology in the field of education?
5.	English Lecturer	46	Master (South China)		5.What is your perspective on the prospect of integrating AI and education?
6.	Lecturer (Department of Information Engineering)	38	Doctorate (Central China)		6.Which behaviors fall under the category of academic integrity, and how do you define them?
7.	Lecturer (Industrial Robotics)	40	Doctorate (Western China)		7.Discuss any academic integrity issues you have encountered personally or learned about from other sources
8.	Lecturer (Computer Science)	35	Master (Western China)		8.How does artificial intelligence technology enhance students' academic performance?
9.	Senior lecturer (Education)	42	Doctorate (East China)		9.To what extent does academic achievement influence the relationship between adopting AI technology and complying with academic integrity?
10.	University Manager	51	Master (South China)		10.What are the potential benefits and challenges of implementing AI technology to enhance students' academic performance? 11.How can educational institutions assist students in conscientiously adhering to the principle of academic integrity while using AI technology?

Source(s): Authors' own work

completing some formulaic tasks, helping with data processing, completing learning tasks, creating more innovative and expressive works, improving teaching effectiveness, meeting students' different learning needs, improving work and teaching efficiency, stimulating students' interest and for more convenient access to academic resources. Literature findings further support these motivations. According to [Alissa and Hamadneh \(2023\)](#), AI technology can serve as a tool to help students enhance their writing abilities and create more original and creative works. AI's capability to accurately translate works into various languages can aid students in expressing themselves effectively and creatively. The identified motivations align with the potential benefits of using AI in educational settings, ranging from enhancing learning experiences and teaching effectiveness to improving efficiency and accessibility to academic resources. By leveraging AI for these purposes,

Figure 2 Motivation and benefits of using AI



educators and students can harness its capabilities to foster innovation, personalization and effectiveness in teaching and learning practices.

Most of the respondents feel motivated to use AI to reduce repetitive labor and improve accuracy. They also believe it can enhance their research and teaching abilities, as well as provide more comprehensive data support and decision-making references. Some respondents highlighted the importance of features such as easy access to question banks, receiving intelligent feedback and deepening subject knowledge. Many participants found AI valuable for providing personalized learning experiences and feedback tailored to individual needs. [Kumar et al. \(2023\)](#) discusses how AI can assist students in tracking previously completed coursework, applying that knowledge to course planning, enabling personalized learning and facilitating review processes. Despite the motivation to use AI, some participants expressed concerns about accuracy and data security risks. While acknowledging the benefits AI offers in terms of efficiency and effectiveness. These individuals also recognized the importance of ensuring that AI systems are reliable and secure, particularly when handling sensitive data and making critical decisions based on AI-generated insights. Addressing these concerns is crucial to building trust in AI technologies and maximizing their potential in educational settings.

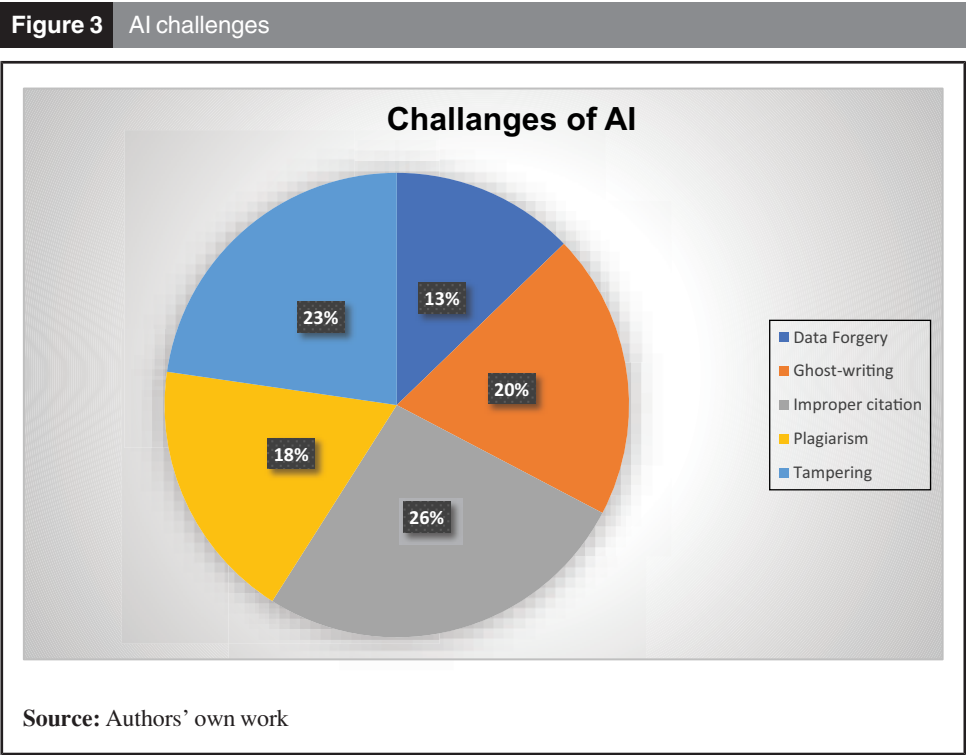
Evaluating the responses for areas being most used for AI, it is found that most of the participants used it for data analysis, model optimization, collecting academic resources, assistance in paper writing, e-learning platform and research for teaching material. Based on our findings, participants leveraged AI to analyze and interpret data to derive insights and make informed decisions in academic research and teaching. Furthermore, AI was employed to optimize models and algorithms, enhancing the efficiency and accuracy of various processes in academic tasks. AI tools were used to gather, organize and access academic resources such as research papers, articles and educational materials more effectively. Participants used AI for assistance in drafting and refining academic papers, proofreading, translation, optimizing writing processes and improving the quality of written content. AI technologies were integrated into e-learning platforms to enhance online teaching and learning experiences, providing personalized learning pathways and interactive educational content. In addition, AI was used in conducting research to identify relevant research and teaching materials, curate educational content and enhance the quality of teaching resources used in academic settings.

The findings align with prior research, as noted by [Pataranutaporn et al. \(2021\)](#), who highlighted that integrating AI into education empowers instructors to tailor activities and lessons to individual students' learning styles, paces and weaknesses. This customization enhances e-learning experiences, optimizes knowledge acquisition and boosts student engagement. In addition, participants reported using AI for creative design, digital art, NLP and other tasks, with applications varying based on their academic focus. For instance, college students used AI to assist them with paper writing and accessing academic resources. In contrast, lecturers used it for model optimization, tutoring material support, student management, teaching evaluation, course planning, code writing, data analysis, machine learning, network security, research and completing e-learning tasks.

Respondents also noted the potential for more effective implementation of AI functions in the future, acknowledging current limitations in power and effectiveness. [Kshetri \(2021\)](#) postulated that AI algorithms are unaffected by task repetition, rendering experience irrelevant and hindering improvement. This underscores the necessity for ongoing advancements in artificial technology to enhance its capabilities. Overall, most participants expressed positive views regarding the impact of artificial technology on academic integrity and performance improvement, as illustrated in [Figure 2](#). The diverse applications of AI in education, coupled with the recognition of its potential for future enhancements, underscore its significance in shaping more personalized, efficient and effective learning environments.

Theme 2: Challenges associated with artificial intelligence integration in academics

Integrating AI into academic settings presents various challenges that need to be addressed to maximize its potential benefits. [Figure 3](#) shows the AI challenges in academics. When considering AI's impact on academic integrity, some respondents noted that unethical practices are less common in lab-based research due to the difficulty of fabricating experimental results with tools like ChatGPT. However, detecting academic dishonesty in homework assignments poses challenges, as plagiarism checks are typically



not conducted. According to [McNabb et al. \(2023\)](#), a significant portion of students confessed to participating in cheating behaviors, including copying from others and submitting work that belonged to someone else. This sheds light on a prevalent issue within academic settings and underscores the importance of addressing concerns about academic integrity.

The study underscores the critical importance of addressing academic integrity issues within college settings to uphold academic quality. Additional insights from respondents further support this notion, highlighting trends such as the use of AI by college students to write papers for assignments, engage in plagiarism detection and even remove plagiarism. Respondents also noted challenges in using AI effectively to evaluate unique characteristics in art and design, leading to concerns about the unethical practice of paper writing with AI. Students use AI for plagiarism detection and further removing it. Respondents mentioned that AI has become complex in identifying and evaluating the unique characteristics of art and design when used in formation. Considering the result, most of the respondents mentioned that AI is constantly being used in the unethical practice of paper writing ([Figure 3](#)).

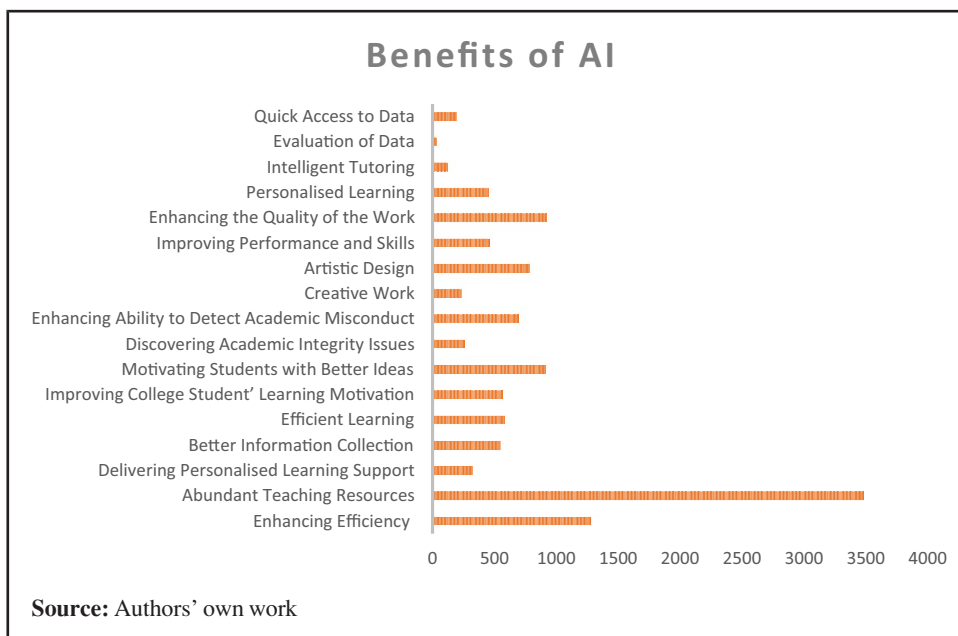
Evaluating the results for which behaviors fall under the category of academic integrity, most respondents mentioned that not including plagiarized work, not reflecting fake results, and not tampering with others' data fall under the academic integrity category. Some participants mentioned that AI should adhere to the principle of integrity while performing any artistic creation. Also, according to [Perkins \(2023\)](#), using AI tools for producing content could result in academic dishonesty and plagiarism or invade academic integrity. Therefore, monitoring the use of AI in educational settings is important.

It has been found that academic integrity serves as the cornerstone of effective learning for higher educational institutional students throughout their academic journeys. One respondent highlighted its indispensable nature in scientific projects and research endeavors. Therefore, the appropriate citation practices and research norms must be adhered to when integrating artificial technology with academic pursuits. However, some participants expressed reservations about this integration in academic settings and raised concerns about various academic issues, including data forgery, ghostwriting, improper citation, plagiarism and tampering. [Gupta \(2016\)](#) has expressed the possibility that students under intense academic pressure may turn to dishonest activities due to AI-driven technologies for identifying plagiarism and automatic essay marking. Similar information has been found from the responses collected from participants, as one of them mentioned that AI would make it more challenging to identify unethical behavior. Therefore, better artificial technology should be developed from the lecture's perspective, which can help identify unethical practices.

Implementing AI in academics has both positive and negative aspects. When participants were asked about it, they mentioned the significant benefits of using AI in academics. These include enhancing efficiency, providing abundant teaching resources, delivering personalized learning support, improving information collection, facilitating efficient learning, boosting college students' learning motivation, inspiring students with innovative ideas, identifying academic integrity issues, enhancing the ability to detect academic misconduct, aiding in creative work and artistic design, improving performance and skills, enhancing work quality, enabling personalized learning, offering intelligent tutoring, assisting tutors in evaluating student data and providing quick access to data ([Figure 4](#)).

According to [Yilmaz and Yilmaz \(2023\)](#), students are interested in being at the center of innovation and creative work and fostering greater imaginative thinking in their learning. We argue that AI is considered the most interesting and fast-growing technology. Users use AI technologies to innovate and develop new projects or find new ways to complete existing ones, aligning with objective two. Furthermore, upon evaluating the responses regarding

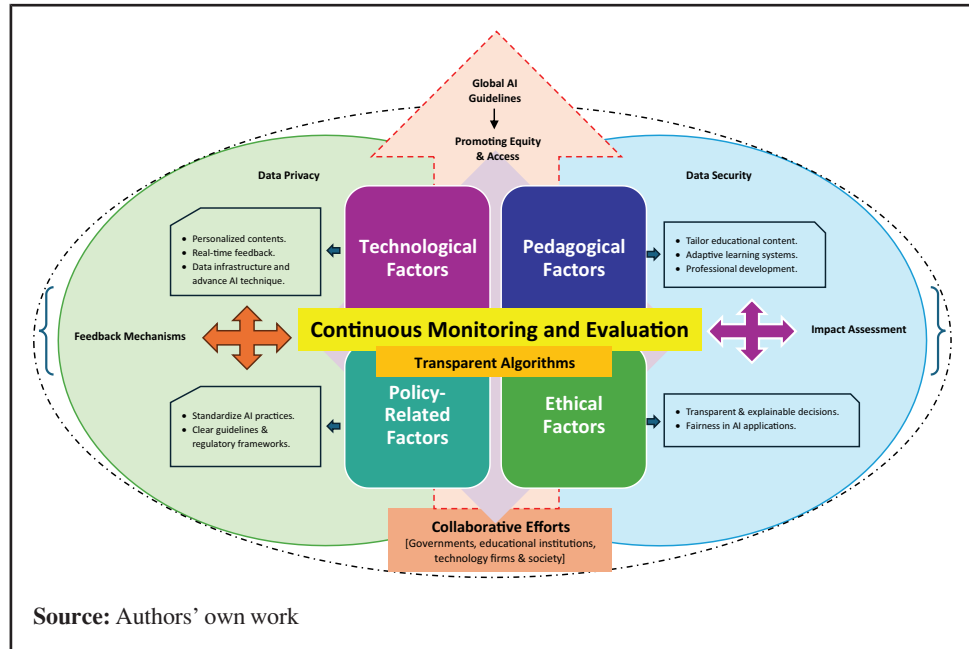
Figure 4 Frequency analysis of motivation



challenges related to the integration of AI in academics, we observed that AI could lead to students overly relying on AI technology for assistance. This has the potential to impede the development of independent thinking among university students. This overreliance can hinder their ability to engage in independent learning and critical thinking. Other issues highlighted include concerns about copyright and creative ethics stemming from the application of technology, potential difficulties students may face in practical applications and problem-solving and issues related to the reliability and interpretability of AI technology. Providing proper guidance to students on the appropriate use of AI without fostering excessive dependence, addressing ethical and data security concerns and ensuring that students do not excessively rely on artificial technology are vital considerations. Overreliance on AI may have negative implications for students' cognitive skills and independent learning capabilities.

Figure 5 presents the framework for responsible AI usage in academic integrity. Our study identifies four main factors that drive the responsible conceptualization and practice of AI in the educational sector. First, the technological factor requires stakeholders to ensure that AI technology can provide personalized content and real-time feedback for teaching, learning and research purposes. Governments and higher educational institutions must develop data structure policies and guidelines and provide advanced AI techniques. Second, the pedagogical factor emphasizes that responsible AI practice involves tailoring content for specific subjects and using adaptive learning systems. To support this, educational institutions should offer professional development programs for instructors. Third, the ethical factor serves as the foundation for clear instructions on using responsible AI. Information should be transparent and accessible to everyone, and decisions regarding the use of AI applications should be explained reasonably. Finally, the policy-related factor addresses the limited access to global standardized AI best practices. In developing countries, many educational institutions lack clear guidelines on how to use responsible AI. AI reports are often used as the main judgment for academic misconduct, which is not always accurate. This leads to instructors penalizing students and researchers based on their own understanding and perception. We argue that clear guidelines and regulatory frameworks can assist educational institutions in making fair judgments and further

Figure 5 Framework of responsible AI usage for academic integrity



investigating plagiarism cases. We also propose that impact mechanisms, assessments, continuous monitoring and evaluation of responsible AI usage in education can strengthen the proposed framework. Data privacy and security are additional concerns that need further discussion in AI infrastructure. The proposed framework may not work well without collaborative efforts among governments, educational institutions, technology firms and society at large.

Discussion

This section discusses the responses to the research question regarding the application of this knowledge in the field of education. Participants indicated that AI could be effectively used in education across various domains, including online teaching platforms, virtual laboratories, intelligent scoring systems, providing competent assistance to students and conducting comprehensive performance evaluations. The versatility of AI in educational settings is evident through its potential applications, such as automated evaluation, facial recognition systems, intelligent online teaching modules, teaching assistance, generating art, image style conversion, creative assistance, implementing intelligent academic management systems and creating virtual learning environments. Moreover, AI can play a pivotal role in enhancing educational activities by facilitating student profiling, intelligent tutoring, automated evaluation processes, intelligent algorithm design, construction of machine learning models and development of intelligent systems. By leveraging AI technologies in education, institutions can enhance teaching methodologies, personalize learning experiences, streamline administrative tasks and create dynamic learning environments that cater to the diverse needs of students. The integration of AI in education holds the promise of revolutionizing traditional educational paradigms and fostering innovative approaches to teaching and learning.

For instance, [Seo et al. \(2021\)](#) suggest that AI holds the promise to transform education, enhancing its accessibility, effectiveness and success for students globally. AI not only aids students in understanding and learning but also provides real-time assistance. Participants generally agreed that when AI is implemented within specified regulations, these

applications align with academic integrity standards. Some respondents specifically noted that technologies like voice recognition and intelligent monitoring could serve as valuable teaching aids. However, a few participants expressed concerns about the reliability of implementing such technologies within the domestic education sector. The potential of AI to revolutionize education lies in its ability to personalize learning experiences, provide immediate feedback and offer tailored support to students. By ensuring that AI applications are ethically developed and effectively regulated, educational institutions can harness the power of AI to enhance teaching and learning practices, ultimately benefiting students on a global scale.

When assessing the responsibility of educational institutions in supporting students to uphold academic integrity while leveraging AI technology conscientiously, respondents emphasized the crucial role of institutes in guiding students on ethical behavior. This guidance is seen as essential for students to make informed decisions and adhere to the principles of academic integrity. Participants suggested that educational institutions should actively educate students on what constitutes ethical behavior and what does not, fostering a culture of integrity within academic settings. By providing clear guidelines and ethical frameworks, institutions can empower students to uphold academic honesty. Moreover, respondents highlighted the importance of institutions in bolstering academic integrity education, enhancing AI detection technologies, preventing instances of academic misconduct and fostering the positive development of students. Strengthening these aspects can contribute to creating a conducive environment where academic integrity is valued and upheld.

Respondents emphasized the importance of implementing improved performance evaluation methods, such as project reports and practical projects, to inspire students to assess and enhance their creativity and practical skills. Encouraging students to engage in reflective practices through these evaluation techniques can motivate them to strive for excellence and personal growth. In addition, participants highlighted the responsibility of educational institutions in guiding college students on how to avoid academic misconduct. This guidance is crucial for fostering a culture of academic integrity within academic settings. This perspective is supported by [Surahman and Wang \(2022\)](#), who suggest that adherence to academic integrity is influenced not only by personal factors but also by institutional conditions. Factors such as the pressure to achieve high scores or a competitive academic environment can contribute to instances of academic dishonesty among students. Therefore, institutions must educate stakeholders on the ethical use of AI in educational settings.

In alignment with the discussions outlined above, instructors play a crucial role in enhancing teaching methodologies, refining their instructional skills and promoting comprehensive student development. By establishing and standardizing effective teaching practices, educators can create an environment conducive to learning and personal growth. This approach not only enhances the educational experience for students but also contributes to their holistic development. Respondents also emphasized the importance of educational institutions in ensuring the ethical and safe implementation of AI technologies in academic settings. This involves enhancing supervision, developing relevant policies and fostering the responsible and healthy integration of AI in education. By addressing ethical concerns and promoting safety measures, institutions can create a secure and supportive environment for leveraging AI technologies in teaching and learning. The parallels between the literature findings and the responses from participants highlight a convergence of perspectives regarding the motivation, benefits and challenges associated with integrating AI in education. Both sources underscore the potential benefits of AI in enhancing educational outcomes, the importance of promoting ethical practices and the need to address challenges related to academic integrity, student development and ethical considerations. This alignment reinforces the significance of fostering a balanced and

responsible approach to leveraging AI in educational settings for the betterment of students and the academic community as a whole.

Limitations and future research

Artificial technology is continuously developing, and this will not remain the same in the future as people use it now. In the future, there is the possibility that AI-related ethical challenges will be solved, and no integrity issue will exist while responsible artificial technology is integrated into academics. However, there is also the chance that it may severely impact the honesty and integrity of the academics with time. Our study has focused on qualitative data to understand AI users' perceptions and has not done much to gain access to quantitative data. Due to limited access to AI literacy and expertise, we have collected data using convenience sampling and not equal probability. Our study has limitations on data analysis using typical statistical analysis. Our findings also deal with uncertainty and acknowledge it as the limitation of this study. AI has different functions, and each academic field might use it for different platforms and purposes. In this study, we did not specifically specify the type of AI and its purpose. Besides that, there is no clear rule on how to use AI in academics. In the future, the University of the Ministry of Education shall set the AI standard to complete the research and assignment. It is also to consider the policy of human-machine interaction by replacing university professors teaching in the classroom. We have discussed AI in general terms. Our study has several limitations that can guide potential future research directions:

- Uncertainty of future AI developments for academics. Our findings acknowledge that AI technology is rapidly evolving, leading to uncertainties about how ethical challenges related to AI integration in academics will be resolved in the future.
- Our study does not explore deeper investigation on AI features types and purposes.

Our study did not specify the types of AI technologies being used or their specific purposes within different academic fields. This lack of specificity limits the depth of understanding regarding the diverse applications of AI in education.

Artificial technology is continuously evolving, and its future applications may differ significantly from current usage patterns. In the years to come, it is plausible that AI-related ethical challenges will be effectively addressed, potentially eliminating integrity issues in the integration of artificial technology in academic settings. However, there remains a possibility that AI advancements could have adverse effects on academic honesty and integrity over time. This uncertainty represents a vital limitation of the current research. AI serves a multitude of functions, and its utilization can vary across different academic disciplines. Unfortunately, our study did not delve into the specifics of AI types and their intended purposes within distinct academic domains. This lack of specificity limits the depth of our understanding regarding the diverse applications of AI in education.

Moreover, a clear set of guidelines on the appropriate use of AI in academic contexts was not established within the scope of our study. As AI technologies continue to shape the educational landscape, it becomes imperative for regulatory bodies such as the university or the Ministry of Education to establish standardized AI protocols to facilitate research and assignments effectively. In addition, considerations regarding the policy implications of human-machine interaction, including the potential replacement of university professors in the classroom, were noted but not extensively explored in our study.

Moving forward, these limitations can inform potential future research directions. Uncertainty of future AI developments. Future studies should address the evolving nature of AI technologies and the implications for academic integrity, considering how ethical challenges may be resolved as AI continues to advance. Exploration of AI features, types and purposes. Our study does not explore the AI features and types. Future studies should

delve deeper into the various types of AI technologies used in different academic fields to gain a comprehensive understanding of their applications and impact. By addressing these limitations and focusing on these potential future research directions, scholars can contribute to a more nuanced understanding of AI's role in academia and develop strategies to ensure its ethical and effective integration in educational settings.

Suggestion for future research

In this study, we conclude that AI will undoubtedly be widely used in the future. With the continuous development and advancement of AI, university stakeholders will have numerous new aspects to identify, evaluate and explore in academic settings. The fundamental AI capabilities include recognizing and matching patterns, decision-making, conceptual linkage and reasoning, implementation and sequential control, planning and problem-solving. These capabilities form the basis of an AI-based smart learning system that encompasses intelligent tutoring and virtual reality. Future research may focus on integrating AI to enhance social and emotional learning in students by recognizing emotions, offering empathetic responses and providing emotional well-being counseling. We suggest that future studies should concentrate on developing clear guidelines or policies for the ethical integration of AI in academia to ensure responsible and effective use of AI technologies in educational environments. In addition, future research should investigate the policy implications of AI in education, particularly regarding human-machine interaction and the potential role of AI in teaching. Exploring these areas can illuminate the impacts of AI on traditional educational practices and the evolving role of educators in the future.

Furthermore, we recommend that future studies explore the development of ethical frameworks for AI usage in academia to tackle challenges related to academic integrity and guarantee that AI technologies are implemented responsibly and transparently. We argue that future research direction can focus on the development of NLP models. It holds immense potential to revolutionize how literature reviews, data searches and academic article summaries are conducted. By leveraging NLP technologies, scholars and students can streamline the process of gathering information, extracting key insights and synthesizing complex academic resources efficiently. The integration of a Lecture-Bot prototype, powered by AI and NLP technologies, can enhance the students' learning experience. The Lecture-Bot can provide interactive lectures, personalized learning paths and real-time feedback to students, promoting engagement and knowledge retention. In conclusion, the fusion of NLP models, AI technologies and innovative educational tools like the Lecture-Bot prototype has the potential to transform traditional learning paradigms, optimize research processes and enhance student achievement through personalized and engaging learning experiences.

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Further reading

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